

**I forced small local
LLMs to answer in JSON.**

**My benchmark lied
to me once.**

Does structured output make
small models worse?
And do they know when they're wrong?

Swipe →

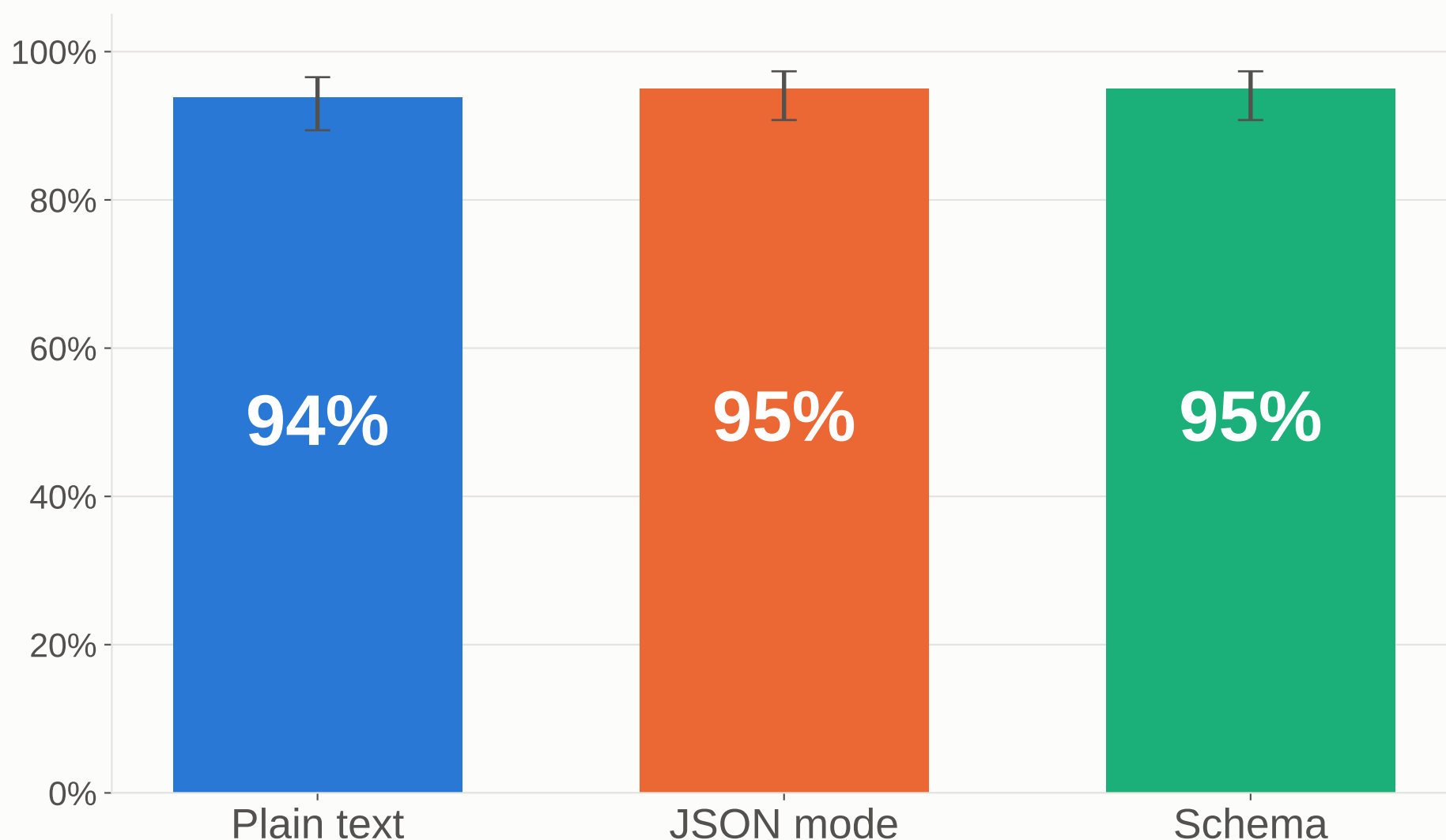
The setup

- 3 open-weight models: Phi-3 3.8B, Mistral 7B, Gemma 2 9B (4-bit, Ollama)
- Same 60 questions about a private document (10 of them unanswerable, to catch hallucination)
- 3 output modes, identical instructions:
plain text · JSON mode · schema-constrained
- 540 answers, paired design, randomised order
- 100% offline on an RTX 4070 laptop. \$0.

**Only the format changes.
Everything else is held constant.**

FINDING 1

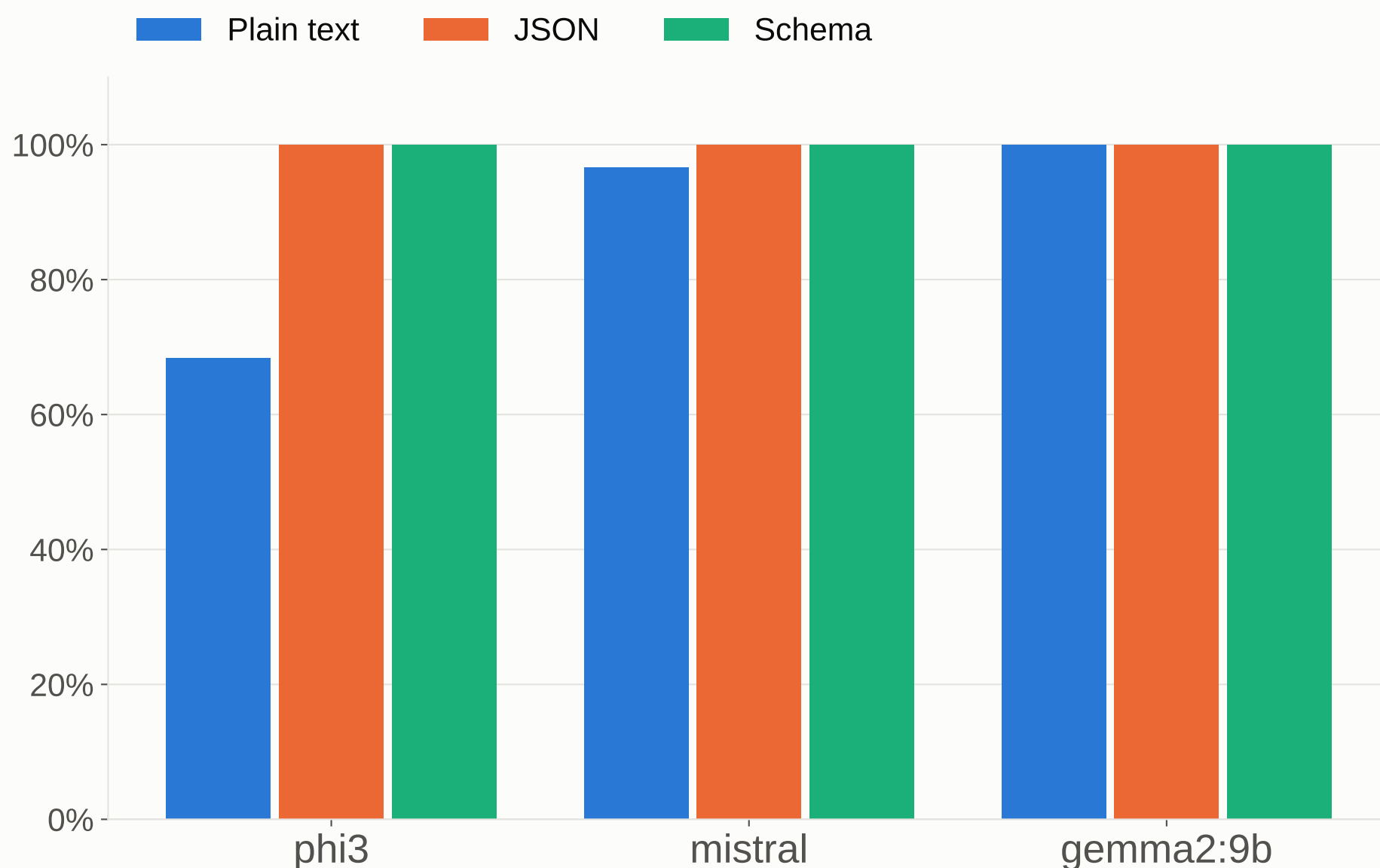
Forcing JSON did not cost accuracy



JSON vs text: +1.1 pp, 95% CI [-1.1, +3.3].
The interval rules out any meaningful drop.

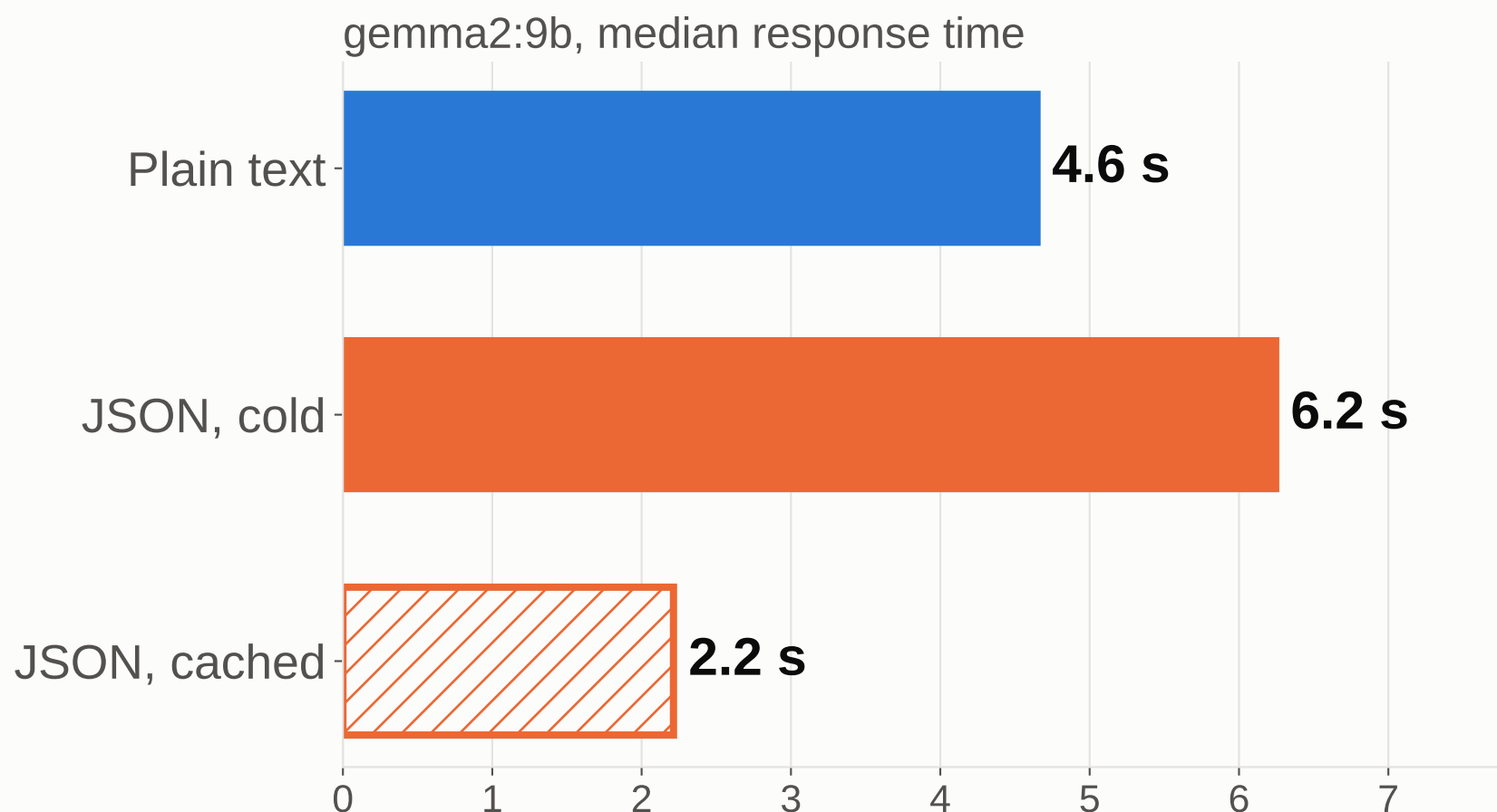
FINDING 2

But it made outputs machine-readable



Valid output on first try: plain text 88%, JSON/schema 100%.
phi3 in plain text: only 68%. It kept dropping the label.

My benchmark said JSON was 18% faster. It wasn't.



JSON and schema prompts were identical, so Ollama's prompt cache gave the second one a free head start. Cold-only: structured output is the same speed or slower. Fixed with a per-request nonce.

FINDING 4

**"100% confident"
barely means anything**

475/540

answers stated confidence = exactly 1.0

- 18 answers were wrong at ≥ 0.9 confidence
- Phi-3 said "33 days" ($28 + 2 = 30$) at 1.0, in all 3 modes
- Confidence vs correctness: AUROC 0.58
(0.5 = coin flip)

Don't route answers to humans on self-reported confidence.

What I'd tell an AI team

1. Use structured output. For grounded Q&A with small models, it cost no measurable accuracy.
2. Measure latency on cold calls. Prompt caching silently fakes speed-ups.
3. Don't trust verbalised confidence. Use retrieval checks or a verifier instead.
4. Report the parser you used. 'Reliability' gains depend on how strict it is.

How it's built

Python · Ollama · Pydantic · NumPy · pandas

Paired bootstrap CIs + McNemar exact tests

Calibration: ECE, Brier, AUROC

No-egress privacy guard, verified in CI

43 tests · interactive results dashboard

Every finding sentence is generated from the data

[github.com/AbdulMuhaiminKhan/
privacy-first-llm-eval](https://github.com/AbdulMuhaiminKhan/privacy-first-llm-eval)

**What would you test next:
newer models, quantisation, or harder reasoning?**